

cal, or pharmaceutical industries. Cognitive manufacturing can optimise process units such as industrial furnaces that are used in metallurgy, mechanical engineering, chemicals, ceramic, and building materials. Furnaces are critical components as these are responsible for around 60% of the energy consumed in the entire plant.

AI and ML hybrid models can take care of combustion optimisation, which is a key to reduce the carbon formation and increase the run-length of the furnaces. Also, the cognitive equipment performance, which is very important in charge gas compressors, boilers, etc on the utility, is important to control them efficiently. We need to ensure their heat recovery is managed well, as it can lead to lower energy losses and help improve carbon emission, which is the primary focus of most industries.

The next step is to bring real-time process optimisation. The critical process parameters can be optimised on a real-time basis to help improve the overall yield of the plant and directly impact the bottom line of the plant. Moreover, there can be vision based solutions for maintaining safety and quality. Lot of things done today by human intervention can all be replaced by vision based solutions.

All these features can come together as a single packaged solution to take care of multiple business use cases, including converting an entire plant into a smarter plant. This approach helps reduce costs and improve yield, uptime, and quality.

In nutshell, the solution involves real-time data that is driving process optimisation and asset performance optimisation on a real-time basis. Real-time data is utilised with the historical data to optimise the processes at the best possible level. This can help reduce the OEE losses.

Another thing it involves is AI, ML, and hybrid products, which will help in production process optimisation by real-time intelligence and practices, and the real-time warning systems to forecast downtime with predictive analytics. By predictive analysis one can configure the process parameters in a target range so that carbon emissions can be reduced.

The aspects discussed so far have



Fig. 3: Incident captured at construction site of a metro station (Courtesy: LivNSense)

a direct impact on operating efficiency by 10-15%, yield improvement by 2-4%, OEE improvement by 8-10%. Green-house gas carbon emissions are reduced by 5% but based on the potential it can reduce the carbon emissions even further by 15%.

Cognitive manufacturing applications can be applied across various types of furnaces, compressors, heat exchanges, boilers, and rotating equipment where failures are very common. If the solution is being taken care of from an end-to-end perspective, the impact can be somewhere between \$20 million to \$100 million a year.

Use cases

Cognitive manufacturing can be applied in a variety of use cases. Let us take an industrial furnace as part of a use case and understand what goes in optimising such a component of the plant. In furnaces, combustion efficiency is responsible for 60-70% energy losses.

Starting from the quality of raw materials being ignored, reduction in the run-length of the furnace is significant. Therefore, making sure that raw

material quality data is available from the laboratory on a continuous basis is the primary aspect. Since industries are unable to provide this data on a real-time basis, a change in management practices is needed.

Next, the analysis of the data performance is done with respect to outcome and identification of the critical process parameter that attributes to impact the overall performance of the furnaces or the plant. Based on that, AI and ML algorithms are designed. Various data points that come from sensors are also analysed. It is important to consider utilities and components like boilers as well.

It is also important to look at dynamic operations, which hold a lot of value at certain times. Let us take an example of temperature measurement. Most of the time in the metal or any such industry, dipstick based manual measurement methods are used. These manual measurements can lead to human error. None of the data analytics can work accurately if there's an error introduced. Therefore, it is necessary to ensure that the data